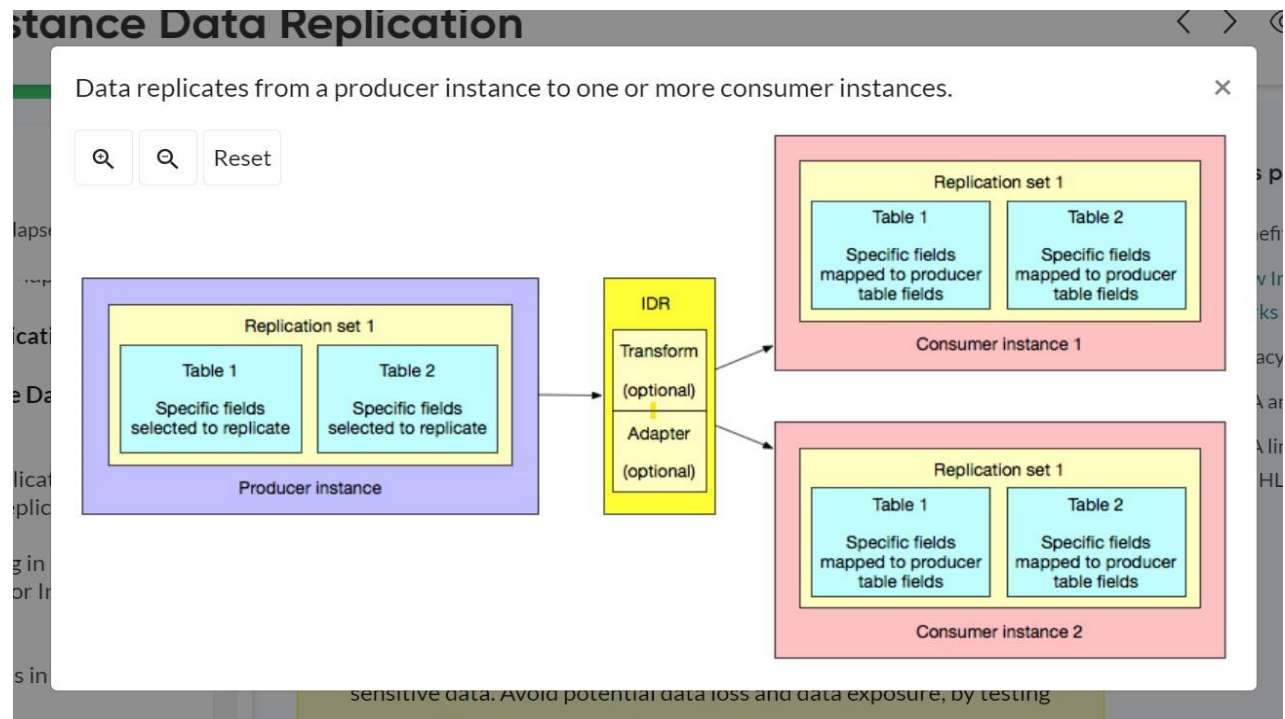


# Instance Data Replication (IDR)

**ServiceNow Instance Data Replication (IDR)** copies data updates from one instance, called the **producer** instance, to one or more other instances called the **consumer** instances.

**IDR** enables you to maintain consistent data across different instances. For example, you can synchronize data between different organizations in your company or even between different companies with separate instances.

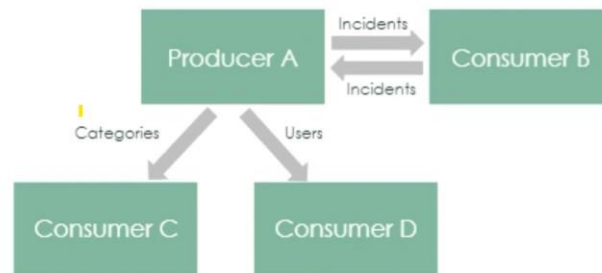
1. **One-to-Many Replication:** Propagates data from a **producer** instance to multiple **consumer** instances.



2. **Bidirectional Replication:** Allows data to flow back and forth between producer and consumer instances.

## Many use cases for IDR

- Single source of truth for foundation data
- Manage data across instances
  - Separate instances for different departments
  - Dev/Production stack
  - External and Internal Instances
  - Managed Services
  - Merger between customers



## Benefits

- Data is automatically replicated to one or more other instances.
- Data can be modified and mapped to any table and table column on other instances. For example, you can modify and map table columns to localize data for different locales.
- Data that is updated on consumer instances can be replicated to the producer instance.
- Data, such as problem requests, can be copied to consumer instances for third parties to use. The third party can update the problem issue on the consumer instance. The data can then be updated on the producer instance.
- Business rules can trigger post-replication workflows, such as generating notifications or validating the replication.
- Data that is in transit during a crash is recoverable.

## Pre-requisites

**The Instance Data Replication (IDR)** plugin requires a separate subscription and must be activated by ServiceNow personnel.

**Roles required:** idr\_admin or admin

**Note:** Detailed information about roles for IDR can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/reference/instance-data-replication-roles.html>

The IDR plugin ID is **com.glide.idr**. (Beginning with the New York release, IDR is supported)

When purchasing an IDR subscription, ServiceNow, Inc. personnel also activate the following plugins that IDR depends upon:

1. **com.snc.db.data\_replicate**
2. **com.glide.transform**
3. **com.glide.kmf**

**Note:** If you want to evaluate the product on a subproduction instance without charge, follow these steps.

URL: <https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/request-instance-data-replication.html>

## Procedure

1. Navigate to All > System Applications > All Available Applications > All.
2. On the All Applications page, select Request Plugin to open the Activate Plugin form on Now Support.

All Applications

Find in Store Request Plugin

FILTERS

▼ Listing type

- ☐ Applications
- ☐ ServiceNow Products

▼ Obtained

- ☐ Installed
- ☐ Not Installed
- ☐ Updates
- ☐ Customized
- ☐ Installation Scheduled

▼ Price

- ☐ Free
- ☐ Paid

▼ License Status

- ☐ Subscription not required
- ☐ Subscribed
- ☐ Subscription unknown

▼ Product family

1466 results

Sort by A - Z

**Action Status Automation**  
Customer Service Management

This plugin tracks blocking records created for tasks and updates the action status indicators on the task list.

Id: com.sn\_action\_status | Paid | by ServiceNow

Install

**Activity formatter**  
Other

Quickly and easily filter the list of activities, or history, on a task form

Id: com.glide.ui\_activity\_formatter | Free | by ServiceNow

Installed

3. On Now Support, select the link to access the Now Support Service Portal Service Catalog.

**Activate Plugin**

In order to enhance the user experience, we have redesigned Activate a Plugin service catalog. Please use the new HI Service Portal 'Activate a Plugin' service catalog item. You can also use Manage Instances page on Service Portal to Activate a Plugin.

Take me to the HI Service Portal Activate a Plugin Service Catalog. 

4. Select your instance.
5. Select Actions > Activate Plugin.
6. On the Activate Plugin form, provide the following information:

| Field                                          | Description                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What is your target instance</b>            | <b>Instance on which to activate the plugin.</b>                                                                                                                                                                                                                                                           |
| <b>Which plugin would you like to activate</b> | <b>Name of the plugin to activate.</b><br><b>Note: If the system does not list the plugin you want or if you are activating the plugin on an OEM or on-premise instance, select the Plugin I'm looking for is not listed check box and then enter the name of the plugin.</b>                              |
| <b>Select Maintenance Date and Time</b>        | <b>The date and time to activate the plugin.</b><br><b>Note: Plugins are activated in two batches, once in the morning and once in the evening, on every business day in the US Pacific time zone. If the plugin must be activated at a specific time, enter the request in the Reason/Comments field.</b> |

## Preparing for instance data replication

### Analyzing table hierarchies

We need to determine if the tables that needs to be replicated is a part of a parent-child table hierarchy.

If the tables are a part of a parent-child table hierarchy, we need to decide whether to replicate the data from the:

1. parent perspective (retaining only columns belonging to the parent table)  
or
2. from the child perspective (retaining all columns that belong to the child tables)

**Note:** To maintain data integrity and ensure that reference fields are populated on the consumer as you'd expect, you must replicate all of the parent and child tables in the table hierarchy.

Detailed information for preserving table hierarchy for IDR can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/preserving-table-hierarchy.html>

# Organizing replication sets

## **Strategy 1**

### **Create a single replication set for each table.**

This option requires more setup, configuration, and time management, but allows you to pause replication for a single table without impacting other tables.

- Replication is performed in parallel using multiple jobs.
- Throughput metrics are easier to track with separate replication sets.
- When an error occurs, replication for a single table is affected instead of all multiple tables.

## **Strategy 2**

### **Create a replication set for a group of related tables.**

This option is easier to set up and manage, but oversight and potential issues impact all tables in the group.

- If you need find information about replication status, the logical name you use for this replication set might be easier to find and understand compared to using distinct replication sets for each table.
- If you want to pause replication, you can do so for the entire group of tables by pausing replication from one replication set.
- If a replication issue occurs on one table and replication stops, this stops replication for all tables in the replication set.

Note: If we decide to group tables in a single replication set, try to limit it to five or fewer tables. If we have more than five tables to replicate, use multiple replication sets with five or fewer tables in each set.

## Deciding which columns to replicate

Avoid including every column in your replication set by default.

For example, if you include a column updated by the system automatically on a frequent basis, IDR may replicate data more often than necessary. When the system replicates this data, it can negatively impact the capacity subscription.

## Preparing target tables on the consumer instance

By default, IDR uses a record's `sys_id` field as the lookup value to keep data synchronized between the producer and consumer instance.

If the target table contains existing data or records from prior data imports, the `sys_id` values in the consumer don't match the `sys_ids` from the producer instance.

For optimal replication results, follow these guidelines:

- Make the producer instance the exclusive data source for the consumer instance.
- Prior to replication, ensure that the target table in the consumer instance is empty. Ideally, the initial records in the target instance are created from data sent exclusively from the producer via a replication set or a clone.
- Ensure that there are no other ongoing data imports or updates to target tables on the consumer instance after replication starts.

**Note:** Alternatively, we can use a custom Coalesce field to identify unique records (instead of the default `sys_id` column) for replication.

Use a custom Coalesce column when records on the consumer instance have a different `sys_id` for the same records on the producer instance:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/idr-coalesce.html>



## Reviewing business rules

1. We can use business rules to trigger workflows after replication, such as sending a notification or validating the replicated data.

<https://docs.servicenow.com/bundle/xanadu-servicenow-platform/page/administer/instance-data-replication/task/post-replication-actions.html>

2. Review any business rules on the consumer instance that might prevent the `idr.system` user from inserting, updating, or deleting records in the target table.

## Reviewing ACLs

Review the ACLs in place on the target table in the consumer instance to ensure that IDR can replicate data successfully. Confirm that the **idr.system** user has the appropriate roles for the target table.

## Reviewing data policies

Review any data policies in place on the target table in the consumer instance to ensure that IDR can replicate data successfully. Confirm that the data you are replicating satisfies the data policy.

**Note:** If failures related to **Business Rules**, **ACLs** and **Data Policies** occur, they appear in the **Instance Data Replication > Replication Payload Error** table on the consumer instance.

View the Error Message field for details on the data policy that is causing the failure.

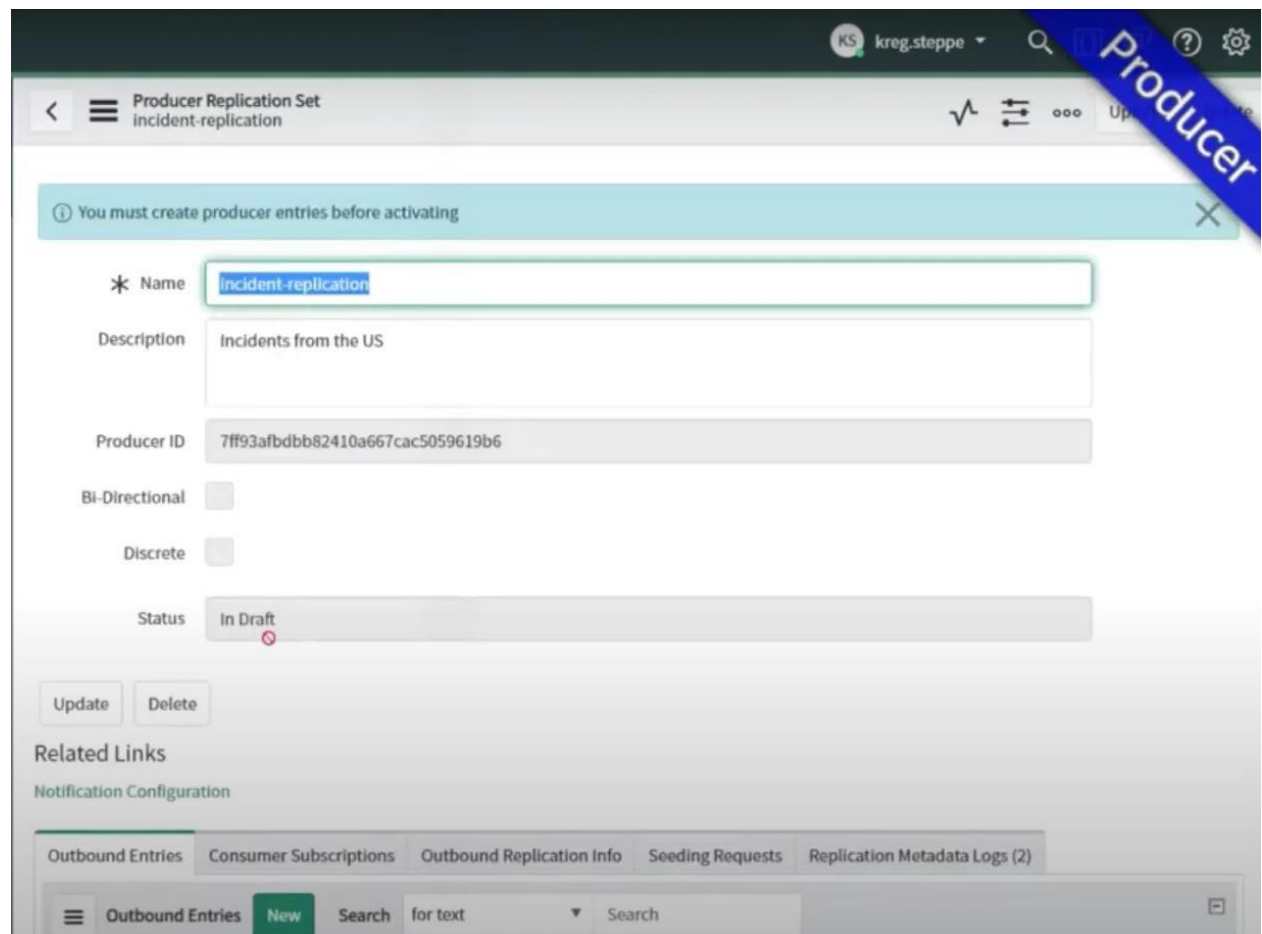
## How Instance Data Replication works

We use the **Instance Data Replication plugin (com.glide.idr)** to replicate data updates on one instance, called the **producer** instance, to one or more other instances, called the **consumer** instances.

By configuring a **producer** replication set, we can specify the tables and table columns on the **producer** instance to replicate.

**Note:** Detailed information about setting up a Producer Replication link can be found here: <https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/create-p-replication-set.html>

### 1. Sample **Producer Replication Set** with status in **In Draft**



The screenshot shows the 'Producer Replication Set' configuration page in ServiceNow. The page title is 'Producer Replication Set incident-replication'. A blue banner in the top right corner says 'Producer'. A light blue message bar at the top states: 'You must create producer entries before activating'. The form contains the following fields:

- Name:** incident-replication
- Description:** Incidents from the US
- Producer ID:** 7ff93afbdbb82410a667cac5059619b6
- Bi-Directional:** ☐
- Discrete:** ☐
- Status:** In Draft (with a red 'x' icon)

Below the form are 'Update' and 'Delete' buttons. Under 'Related Links', there is a link for 'Notification Configuration'. At the bottom, there are tabs for 'Outbound Entries', 'Consumer Subscriptions', 'Outbound Replication Info', 'Seeding Requests', and 'Replication Metadata Logs (2)'. The 'Outbound Entries' tab is active, showing a 'New' button and a search bar.

2. Click on **New** option under **Outbound Entries** to select the table that needs to be exported. Here we have the option to select the table and :

- set Filter conditions
- Option to include attachments
- Include Fields

Replication Entry  
New record  
[Producer\_replication\_entry view\*]

\* Source Table Name: Incident [incident]

Filter: Add Filter Condition Add \*OR\* Clause

-- choose field -- -- oper -- -- value --

Include Attachments ☐

Included Fields

| Available           |  | Selected           |
|---------------------|--|--------------------|
| Active              |  | Sys ID (Mandatory) |
| Activity due        |  |                    |
| Actual end          |  |                    |
| Actual start        |  |                    |
| Additional assigne  |  |                    |
| Additional comme    |  |                    |
| Approval            |  |                    |
| Approval history    |  |                    |
| Approval set        |  |                    |
| Assigned to         |  |                    |
| Assignment group    |  |                    |
| Business duration   |  |                    |
| Business resolve ti |  |                    |
| Caller              |  |                    |
| Category            |  |                    |
| Caused by Change    |  |                    |
| Change Request      |  |                    |

Note: Edge Encrypted fields, Password (2-Way Encrypted) Type fields are not available for replication.

3. Once an **Outbound Entries** is created, we have the option to **activate** the **Producer Replication Set**

To exit full screen, press **Esc**

KS kreg.steppe

Producer Replication Set  
incident-replication

Activate Update

Producer ID: 7ff93afbdbb82410a667cac5059619b6

Bi-Directional ☐

Discrete ☐

Status: In Draft

Activate Update Delete

Related Links

Notification Configuration

Outbound Entries (1) Consumer Subscriptions Outbound Replication Info Seeding Requests Replication Metadata Logs (2)

Outbound Entries New Search for text Search 1 to 1 of 1

Replication Entries

| Source Table Name | Filter | Include Attachments | Include fields                           | Entry Set            |
|-------------------|--------|---------------------|------------------------------------------|----------------------|
| incident          | false  |                     | parent, made_sla, caused_by, watch_list, | incident-replication |

4. Producer Replication Set in Active Replication status

To exit full screen, press **Esc**

KS kreg.steppe

Producer Replication Set  
incident-replication

Deactivate Update

Name: incident-replication

Description: Incidents from the US

Producer ID: 7ff93afbdbb82410a667cac5059619b6

Bi-Directional ☐

Discrete ☐

Status: Active Replication

Deactivate Update Delete

Related Links

Generate New Shared Key  
Replication Setup Instructions  
Notification Configuration

5. Configuring a sample **Consumer Replication Set** in Consumer instance, once created the status will be **Approval Pending**

Detailed information about creating a **Consumer Replication Set** can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow->

[platform/page/administer/instance-data-replication/task/create-c-replication-set.html](https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/create-c-replication-set.html)

To exit full screen, press **Esc**

KS kreg.steppe

Consumer Replication Set  
Incident-replication

\* Producer Replication Set Name: Incident-replication

Description:

Producer Instance URL: https://kproducer.service-now.com/

Producer ID:

Producer Set ID:

Bi-Directional: ☐

Discrete: ☐

Status: In Draft

Consumer Approval Status: Approval Pending

Request Producer Approval Update Delete

Related Links

6. Then go back to the **Producer Replication Set > Consumer Subscriptions** to approve the request.

Detailed information about granting access to replicated data for IDR can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow->

[platform/page/administer/instance-data-replication/task/approve-consumer.html](https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/approve-consumer.html)

To exit full screen, press **Esc**

KS kreg.steppe

**Producer**

### Producer Replication Set

incident-replication

Product ID: 1172d1a2b98274v000f1b1a2723012300

BI-Directional: ☐

Discrete: ☐

Status: Active Replication

Deactivate Update Delete

#### Related Links

[Generate New Shared Key](#)  
[Replication Setup Instructions](#)  
[Notification Configuration](#)

Outbound Entries (1) Consumer Subscriptions (1) Outbound Replication Info Seeding Requests Replication Metadata Logs (4)

Show / hide filter Consumer Subscriptions Search Consumer Instance ID Search 1 to 1 of 1

#### Replication Subscriptions

|  | Consumer Instance ID             | Consumer Instance Name | Consumer Approval Status | Consumer Status |
|--|----------------------------------|------------------------|--------------------------|-----------------|
|  | 035abe3bdb30a410f37acac505961958 | kconsumer              | Approval Pending         | In Draft        |

KS kreg.steppe

**Producer**

### Replication Subscription

kconsumer

Update Approve

Producer Set: incident-replication

Consumer Instance ID: 035abe3bdb30a410f37acac50596

Consumer Instance Name: kconsumer

Consumer Status: In Draft

Consumer Status Message:

Consumer Approval Status: Approval Pending

Last Heartbeat Sent:

Last Heartbeat Received:

Last Data Message Lag:

Last Message Network Lag:

Is Consumer Remote: ☐

Update Approve

#### Seeding Requests

Seeding Requests Search Created Search

Subscription = kconsumer

| Created | Status | Start Time | Run Duration | Percent Complete | End Time |
|---------|--------|------------|--------------|------------------|----------|
|---------|--------|------------|--------------|------------------|----------|

7. Click on **YES**



Once approved, the **Producer ID** and **Producer Set ID** fields will be updated in **Consumer Replication Set**

A screenshot of a web application interface showing the "Consumer Replication Set" form. The form is titled "Consumer Replication Set" and "Incident-replication". It includes a "Request Producer Approval" button and an "Update" button. The form fields are: "Producer Replication Set Name" (with a dropdown menu showing "Incident-replication"), "Description" (text area), "Producer Instance URL" (text field with "https://kproducer.service-now.com/"), "Producer ID" (text field with "7ff93afbdbb82410a667cac5059619b6"), "Producer Set ID" (text field with "6805d82adb096410067a9eb5db961960"), "BI-Directional" (checkbox), "Discrete" (checkbox), "Status" (dropdown menu showing "In Draft"), and "Consumer Approval Status" (dropdown menu showing "Approved"). At the bottom, there are buttons for "Request Producer Approval", "Update", and "Delete". A green diagonal banner with the word "Consumer" is overlaid on the right side of the form. The top of the page has a navigation bar with a search icon, a user profile icon, and a settings icon.

8. We can then activate the Consumer Replication Set

The screenshot shows the 'Consumer Replication Set' configuration page for 'incident-replication'. The status is 'In Draft'. A green diagonal banner with the word 'Consumer' is in the top right corner.

| Field                    | Value                              |
|--------------------------|------------------------------------|
| Replication Set Name     | incident-replication               |
| Description              |                                    |
| Producer Instance URL    | https://kproducer.service-now.com/ |
| Producer ID              | 7ff93afbdbb82410a667cac5059619b6   |
| Producer Set ID          | 6805d82adb096410067a9eb5db961960   |
| Producer Name            | kproducer                          |
| Bi-Directional           | <input type="checkbox"/>           |
| Discrete                 | <input type="checkbox"/>           |
| Status                   | In Draft                           |
| Consumer Approval Status | Approved                           |

Buttons: Activate, Update, Delete

The status will get updated to **Active Replication**

The screenshot shows the 'Consumer Replication Set' configuration page for 'incident-replication'. The status is now 'Active Replication'. A green diagonal banner with the word 'Consumer' is in the top right corner.

| Field                           | Value                              |
|---------------------------------|------------------------------------|
| * Producer Replication Set Name | incident-replication               |
| Description                     |                                    |
| Producer Instance URL           | https://kproducer.service-now.com/ |
| Producer ID                     | 7ff93afbdbb82410a667cac5059619b6   |
| Producer Set ID                 | 6805d82adb096410067a9eb5db961960   |
| Producer Name                   | kproducer                          |
| Bi-Directional                  | <input type="checkbox"/>           |
| Discrete                        | <input type="checkbox"/>           |
| Status                          | Active Replication                 |
| Consumer Approval Status        | Approved                           |

Buttons: Update, Pause, Delete



9. In the **Producer Replication Set**, we can seed the Table

The screenshot shows the 'Consumer Replication Set' interface for 'Incident-replication'. The status is 'Active Replication' and the consumer approval status is 'Approved'. There are buttons for 'Update', 'Pause', and 'Delete'. Below this, there are 'Related Links' for 'Synchronize Replication Entries', 'Track in Update Set', and 'Notification Configuration'. The main section has tabs for 'Inbound Entries (1)', 'Inbound Replication Info (1)', 'Seeding Requests', and 'Replication Metadata Logs (13)'. The 'Inbound Entries' tab is active, showing a table with columns: 'Source Table Name', 'Filter', 'Include Attachments', 'Include fields', and 'Entry Set'. A tooltip points to the 'Seed' button, stating: 'Request all records matching the filter criteria on the selected tables from the producer instance'. The 'Seed' button is highlighted in blue.

Consumer

KS kreg\_steppe

Consumer Replication Set  
Incident-replication

Discrete ☐

Status Active Replication

Consumer Approval Status Approved

Update Pause Delete

Related Links

[Synchronize Replication Entries](#)  
[Track in Update Set](#)  
[Notification Configuration](#)

Inbound Entries (1) Inbound Replication Info (1) Seeding Requests Replication Metadata Logs (13)

Inbound Entries Search for text Search 1 to 1 of 1

Consumer Replication Entries

Source Table Name Filter Include Attachments Include fields Entry Set

Request all records matching the filter criteria on the selected tables from the producer instance

Seed Seed With History Actions on selected rows...

parent, made\_sla, caused\_by, watch\_list, Incident-replication

Status of the Seeding Request

The screenshot shows the 'Producer Replication Set' interface for 'Incident-replication'. The status is 'Active Replication'. There are buttons for 'Deactivate', 'Update', and 'Delete'. Below this, there are 'Related Links' for 'Generate New Shared Key', 'Replication Setup Instructions', and 'Notification Configuration'. The main section has tabs for 'Outbound Entries (1)', 'Consumer Subscriptions (1)', 'Outbound Replication Info (1)', 'Seeding Requests (1)', and 'Replication Metadata Logs (15)'. The 'Seeding Requests' tab is active, showing a table with columns: 'Consumer Instance Name', 'Created', 'Status', 'Start Time', 'Run Duration', and 'Percent Complete'. The table shows one entry for 'kconsumer' with a status of 'Completed' and 100% completion.

Producer

KS kreg\_steppe

Producer Replication Set  
Incident-replication

Discrete ☐

Status Active Replication

Deactivate Update Delete

Related Links

[Generate New Shared Key](#)  
[Replication Setup Instructions](#)  
[Notification Configuration](#)

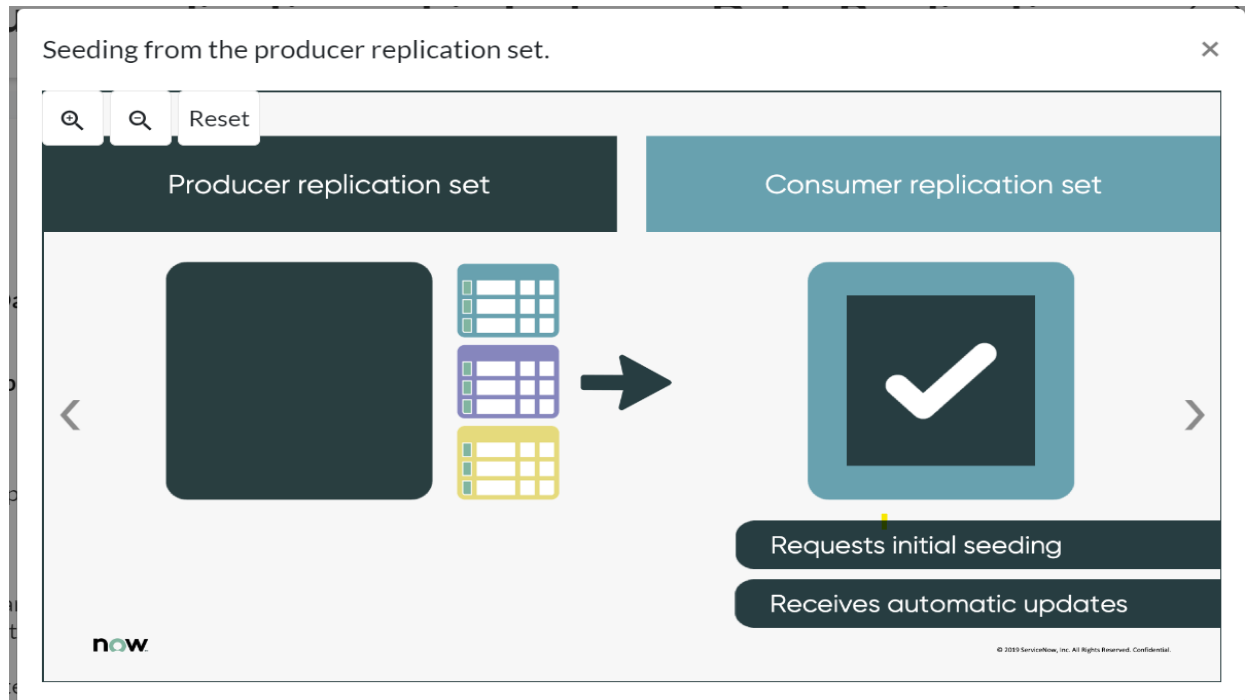
Outbound Entries (1) Consumer Subscriptions (1) Outbound Replication Info (1) Seeding Requests (1) Replication Metadata Logs (15)

Seeding Requests Search Subscription Consumer Instance Name Search 1 to 1 of 1

Replication Set = Incident-replication

|                          | Consumer Instance Name | Created             | Status    | Start Time          | Run Duration | Percent Complete |
|--------------------------|------------------------|---------------------|-----------|---------------------|--------------|------------------|
| <input type="checkbox"/> | kconsumer              | 2020-12-15 08:29:46 | Completed | 2020-12-15 08:29:59 | 1 Second     | 100%             |

**Seeding:** Syncing the **producer** and **consumer** replication sets requires that you do a one-time download (called **seeding**) of all the **producer** replication set data to the **consumer** instances.



You can initiate **seeding** requests on a **consumer** instance when you activate a **consumer** replication set. After **seeding**, replication involves data updates only. An audit trail contains a history of those record updates.

**Note:** Detailed information about **seeding** can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/seed-consumer-instance.html>

## Transform replication data for Instance Data Replication

Used to Transform **producer** instance data for tables or table columns that are named differently on **consumer** instances in Instance Data Replication (IDR).

By default, the table data on a **producer** instance goes into the tables of the same name on **consumer** instances. **Transformation** is the process of replicating producer data in tables or table columns that have a different name on the **consumer** instances.

Note: Detailed information can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/modify-replicated-data.html>

## Adapters

Adapters modify the **producer** data before inserting the data on **consumer** instances in Instance Data Replication (IDR).

Adapters perform string and mathematical operations, such as converting one currency to another, or converting one time zone to another.

Each adapter has **Name** and **Description** fields. The name appears in the Adapter column. Use the **Description** field to explain the purpose of the data conversion.

**Note:** Detailed information can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/reference/adapter-descriptions.html>

## Steps to Revoke access to replicated data for Instance Data Replication

can be found here: <https://docs.servicenow.com/bundle/xanadu-servicenow-platform/page/administer/instance-data-replication/task/revoke-consumer-sets.html>

### Monitoring Instance Data Replication

You can monitor the status of active producer and consumer replication sets, scheduled jobs, seeding requests, and license or usage details through the Instance Data

Replication (IDR) Monitoring Dashboard.

### Accessing the IDR Monitoring dashboard

Users with the admin or idr\_admin role can access the dashboard. Access the IDR Monitoring dashboard in the following way:

Navigate to Instance Data Replication > Monitoring Dashboard.

Choose either: IDR Overall Monitoring or IDR License and Usage Details dashboard.

The IDR Overall Monitoring dashboard monitors the following:

- Active Producer Replication Sets
- Active Consumer Replications Sets
- Suite of IDR Scheduled Jobs
- Producer Seeding Requests Within Last 7 Days
- Consumer Seeding Requests Within Last 7 Days

**Note:** Detailed information about the IDR Dashboard can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/instance-data-replication-dashboard.html>

## Upgrading legacy replication sets to V2 in Instance Data Replication

1. Upgrade Instance Data Replication (IDR) legacy replication sets to V2 for improved message processing and replication performance. It utilizes **Hermes Messaging Service** for **Kafka** message transport. Offering **faster**, **scalable**, and **reliable** data replication

The Hermes Messaging Service is a Now Platform capability that's automatically enabled when the IDR plugin (com.glide.idr) is activated.

2. Upgrading replication sets from legacy to V2 is a seamless process. Active data replication from the producer to consumer continues throughout the upgrade, which means you don't need to pause replication before you begin. Replication continues using the V2 replication sets after the upgrade is finished.

**Note:** Data sent from a V2 producer replication set is retained in the message queue for 24 hours.

Detailed Information about upgrading legacy replication sets to V2 in IDR can be found here:

[https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/upgrading-legacy-replication-sets-v2.html#upgrading-legacy-replication-sets-v2\\_section\\_r2w\\_4b5\\_lvb](https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/upgrading-legacy-replication-sets-v2.html#upgrading-legacy-replication-sets-v2_section_r2w_4b5_lvb)

### Upgrade Eligibility:

- Both producer and consumer instances must be on **Utah** or higher.
- Some configurations may render certain legacy sets **not eligible**.

Note that some Utah legacy replication sets aren't eligible for upgrade to V2, depending on their configuration (for example, the replication set can't be

bidirectional or discrete)

- Check eligibility via the **Upgrade Eligibility** column in the Consumer Subscriptions list.

## Compatibility:

If you upgrade a producer replication set to V2, you must upgrade the subscribed consumer replication sets to V2 as well to take advantage of the V2 processing improvements.

If you don't upgrade the consumers to V2, they continue to consume messages from the legacy producer replication set.

## Upgrade Steps:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/upgrade-legacy-replication-sets.html>

## Revoke access to replicated data for Instance Data Replication

Revoke a consumer's access to replicated data if you believe that consumer instance should no longer receive data in Instance Data Replication (IDR).

When you generate a new, shared encryption key, approved consumer sets automatically receive the new key and data replication is not interrupted. Any consumer set with a status of Denied or Pending Approval does not receive the new encryption key, and replication ceases.

## Procedure

1. On the producer instance, navigate to Instance Data Replication > Producer Replication Sets.
2. Select the producer replication set that has consumers whose access needs to be revoked.

The instance identifies the consumers under the Consumer Subscriptions tab.

Related Links

[Generate New Shared Key](#)

[Replication Setup Instructions](#)

[Track in Update Set](#)

[Notification Configuration](#)

Outbound Entries (1)

Consumer Subscriptions (1)

Outbound Replication Info (1)

Seeding Requests

Replication Metadata Logs (13)

Consumer Instance |

Actions on selected rows...

Replication Subscriptions

| <input type="checkbox"/> | Consumer Instance ID             | Consumer Instance Name | Consumer Approval Status | Consumer Status | Consumer Status Message | Is Consumer Remote |
|--------------------------|----------------------------------|------------------------|--------------------------|-----------------|-------------------------|--------------------|
| <input type="checkbox"/> | 88b6e346dbc305140ae3ac44d49619cd | idrtesting100003       | Approved                 | In Draft        |                         | true               |
| 1 to 1 of 1              |                                  |                        |                          |                 |                         |                    |

3. On the Consumer Subscriptions tab, select the option for the consumer instance whose permissions you want to revoke.
4. In the Actions on selected rows list, select Revoke.

Instance Data Replication generates a new encryption key and shares it with all approved consumer sets. All other consumer sets stop receiving replication data. Admins of those consumer replication sets must [reapply to restore access to the replication data](#).

## **IDR limitations and when not to use IDR**

1. IDR runs in ServiceNow data centers only, which means there is no support for using IDR with on-premise instances.
2. Do not use IDR to clone instances.

IDR does not replicate metadata tables, child metadata tables, and most user and system tables. IDR is designed to replicate data, not clone instances.

### **3. Limitations for Replicating the updated Records:**

- Maximum size of records is 32MB
- IDR supports replicating approximately 1 Million records per day

### **4. Limitation for Seeding Records:**

- Initial seeding of the tables must not exceed 3 million records per replication set.
- Replication seeding must not take longer than 7 days to complete.
- Producer and consumer instances must be in the same geographic region.

5. Avoid replicating system tables (tables with a sys\_ prefix). If you replicate system tables (like sys\_user, sys\_user\_group, or sys\_user\_grmember) that have existing data in the consumer instance, insert and update failures can occur during replication

For example the Application File [sys\_metadata] table and tables that extend [sys\_metadata] (including the Business Rules [sys\_script], Catalog [sc\_catalog], and Workflow [wf\_workflow] tables) are excluded and can't be replicated.

6. Avoid replicating CMDB tables. If you determine the CMDB tables must be replicated, you must use conditions to constrain the count of replicated records and ensure all required columns are included in the replication set.
7. You can't replicate certain tables in Instance Data Replication (IDR). Child tables of tables in the exclusion list are also excluded.

### **Excluded tables**

These tables are excluded from IDR:

- Tables with the prefixes V\_PREFIX, SYSX\_, sys, ts\_, ua\_, usageanalytics\_, wf\_, ecc\_, clone\_, jrobin\_, rrd\_, imp\_, pf\_, pfd\_, idr\_

sys\_domain is an exception.

- Rotated tables, including syslog, sys\_querystat, ecc\_queue, ecc\_event, cmdb\_metric, sysevent
- Tables without a sys\_id
- sso\_federation
- saml2\_update1\_properties
- sso\_properties
- digest\_properties
- Instance

Child tables that extend these tables are also excluded from IDR and should not be included in replication sets because replication fails if they are included. For example, sc\_cat\_item extends sys\_metadata, which is an excluded table. This makes sc\_cat\_item an excluded table due to inheritance.

Tables to avoid replicating:

- cmdb
- cmdb\_ci



If you determine the CMDB tables must be replicated, you must use conditions to constrain the count of replicated records and ensure all required columns are included in the replication set.

## **8. Attachments**

Avoid using HLA to replicate a series of large attachments on a regular basis. If you need to include attachments that are larger than 10 MB regularly, monitor HLA to ensure that the lag time doesn't exceed expectations.

9. You cannot replicate Edge Encrypted, column level encryption (CLE), and Password (2-Way Encrypted) fields.

## **Additional Information:**

1. Information about monitoring data replication activities and troubleshoot potential issues in Instance Data Replication (HLA) can be found here:

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/administering-instance-data-replication.html>

2. Information about resolving data replication errors in Instance Data Replication

<https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/reference/common-issues-idr.html>